

ROCKY MOUNTAIN UNIVERSITY

Automated Registrar's Office Workflow

Capstone Project | University of Utah | 2026

Agenda

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Team, background, problem, objectives & approach

02

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Deliverables, site visit insights, process transformation

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04

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05

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Recommendations and path forward

Project Overview

Context, problem space, and objectives

Meet the Team

Capstone Students

Sooyoung Lee

Project Management &
Frontline User Education

Paul Arnold

Requirements Gathering &
User Acceptance Testing

Robert Bost

Workflow &
Automation Developer

RMU Project Sponsors & Stakeholders

Jerry Van Wie

RMU IT Project Manager

Daniel Vela

RMU IT Analyst

Registrar's Office

Multiple RMU Staff

Background

Institution

Rocky Mountain University of Health Professions is a graduate university specializing in health professions education.

Current Process

The Registrar's Office manually monitors weekly student attendance across all programs and courses.

Data Sources

Attendance data is collected from multiple systems (Canvas LMS, SIS) and scattered spreadsheets each week.

Problem Statement

1

Manual Data Cleaning

Hours spent every week converting, validating, and enriching attendance CSVs by hand.

2

Multiple Systems

Staff must jump between Canvas, SIS, spreadsheets, chat, and email to complete a single review cycle.

3

Repetitive Work

The same sequence of checks and notifications is performed weekly with little variation.

4

Human Error Risk

Manual processes introduce risk of missed flags, incorrect warnings, or incomplete audit trails.

Business Impact

Administrative Workload

Significant time consumed by repetitive, low-value tasks every week.

Delayed Notifications

Students receive warnings later than ideal, reducing intervention effectiveness.

Staff Inefficiency

Highly skilled Registrar staff spend hours on data prep instead of student support.

Scalability Limits

Manual process cannot scale as enrollment and program complexity grow.

Project Objectives & Goal

Objectives

- 1 Automate the weekly attendance review workflow
- 2 Dramatically reduce manual data cleaning effort
- 3 Standardize student notification language and timing

GOAL

Improve efficiency, accuracy, and consistency through intelligent workflow automation

Reduce repetitive work so Registrar staff can focus on high-value student support and exception handling.

Approach to Completing the Project

01

Stakeholder Interviews

Deep conversations with Registrar staff, Student Affairs, and IT to surface real pain points and edge cases.

02

Gemba Walk

On-site observation of the actual weekly attendance review process to understand the true workflow.

03

SDLC Methodology

Structured requirements → design → development → testing → training cycle with iterative feedback.

04

Microsoft Power Platform

Leveraged Power Automate, SharePoint, and related services for a low-code, maintainable solution.

Results

What we delivered and observed

Deliverables

SOW Deliverable	Project Outcome
Workflow Analysis	Documented As-Is and To-Be registrar workflows
System Architecture	Designed data flow between Data Warehouse, SharePoint, and Power Automate
Automation Pipeline	Developed an automated attendance processing workflow
SharePoint Solution	Built review and approval interface for Registrar staff
Testing & Validation	Compared automated results against the manual process
Documentation & Training	Created technical documentation and user guidance

Site Visit Goals

01

Understand the Why

Discover the Registrar Office's true purpose and constraints behind the weekly attendance process.

02

Observe the Process

Watch the full attendance review cycle end-to-end to identify friction points and workarounds.

03

Build Relationships

Establish trust with frontline staff so feedback is honest and solution adoption is smoother.

University Policy & Federal Regulation

KEY REQUIREMENT

“...the University is required to demonstrate that students are participating on a weekly basis in each course in which they are enrolled.”

This regulatory obligation drives the entire weekly attendance monitoring process.

Site Visit Observations

Manual Data Cleaning

- ▶ Convert date formats for sorting
- ▶ Add missing section numbers from SIS
- ▶ Fill in course type (full vs partial semester)
- ▶ Verify exclusion criteria (block, holiday, on-site, NDS)
- ▶ Manually check Canvas for non-CSV participation signals

Exception & Notification Path

- ▶ Check for known exceptions via chat & Student Affairs
- ▶ First Email: Friendly Reminder
- ▶ Second Email: Final Warning
- ▶ Administrative Withdrawal process
- ▶ Create complete audit trail in the SIS

Planning & Requirement Analysis: AS-IS

Canvas Team

Generate Weekly Activity CSV &
Email to Registrar's Office

Registrar's Office

Clean CSV: dates, sections,
course type, exclusions

Manually check Canvas
access overview for participation

Post flagged students on RO Chat
& discuss exceptions

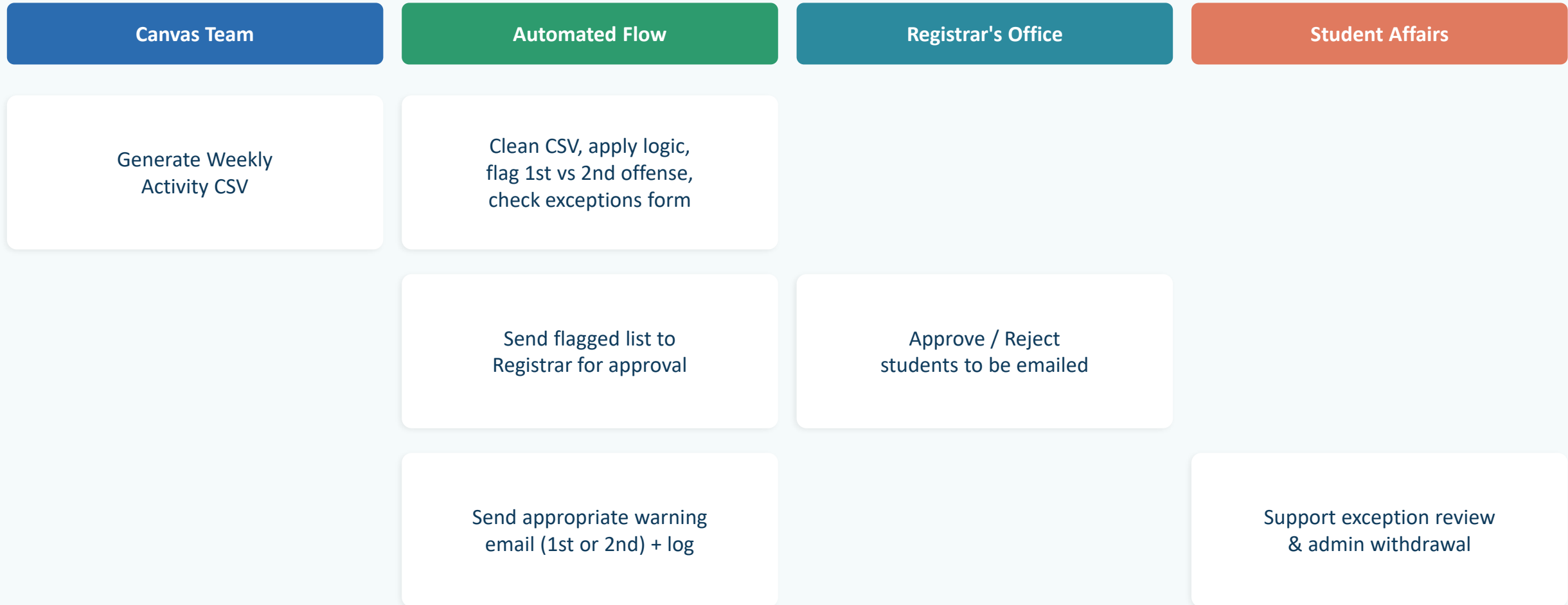
Issue 1st or 2nd warning email
based on history (14-day window)

Student Affairs

Verify known student issues
or exceptions

Review & confirm administrative
withdrawal if needed

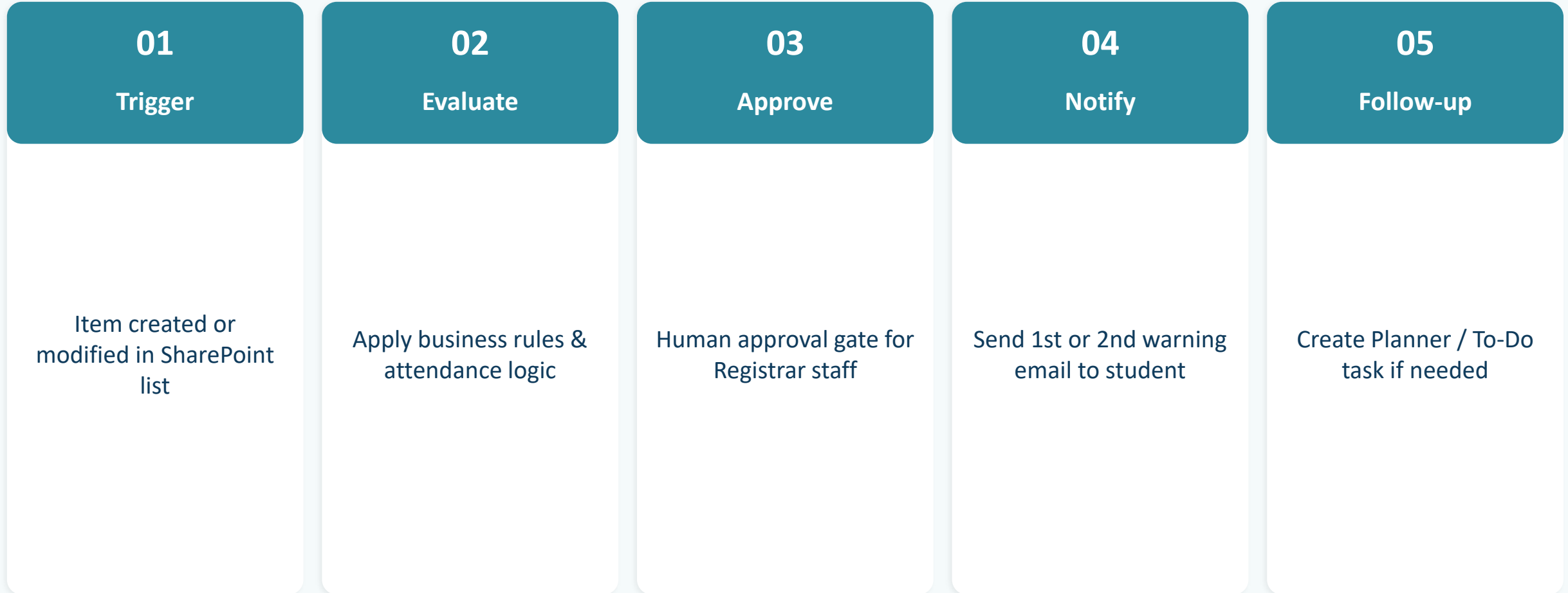
Planning & Requirement Analysis: To-Be



Design & Development

Architecture and automation pipeline

End-to-End Flow



Implementation Outline — Main Flow

SharePoint Trigger → Approval → Notification

1 Trigger

When an item is created or modified in the SharePoint list

2 Get Item

Retrieve full SharePoint item details for processing

3 Condition

Apply business rules to determine first-warning need

4 Compose + Update

Prepare data and write status back to the list

5 Start & Wait Approval

Human approval gate for Registrar staff review

6 Branch

Condition on Approve / Reject decision

7 Send Email

Compose subject/body → Send email (V2) to student

8 Status Marker

Write “First Warning Sent” status (insertion point for follow-up)

Implementation Outline — Follow-up Scope

After First Warning Sent → Task Creation & Error Handling

A Initialize Variables

Capture studentID & staffID from the Get item action

B Time Calculations

Current time → Add to time → calculate dueDate

C Compose Subject

Build dynamic subject line for the Planner / To-Do request

D Compose Note

Create task description / note body with context

E Compose Request Body

Assemble JSON payload for the HTTP call

F HTTP POST

Call Planner task / To-Do / Teams endpoint

G Success Branch

Log debug information and record success

H Failure Branch

Send failure notice + robust error handling

Technical Highlights

SharePoint as System of Record

Central list holds flagged students, status, and approval history for full auditability.

Human-in-the-Loop Approval

Power Automate approval action ensures Registrar staff retain final control before any email is sent.

Dynamic Email Templates

First-warning and second-warning messages are composed with the correct tone and content based on offense count.

Exception Form Integration

Shared form data is checked automatically so known exceptions never generate false positives.

Planner / To-Do Follow-up

After first warning, a task is created so staff can track resolution without losing context.

Robust Error Handling

Dedicated failure branch with notifications prevents silent failures in the automation pipeline.

Lessons Learned

What we discovered along the way

Challenges & How We Overcame Them

Scope Clarity

Early ambiguity around exact business rules and edge cases. We overcame this by investing heavy time in stakeholder interviews and a formal Gemba walk before writing any code.

User Criteria Definition

What counts as “participation” is nuanced (downloads, messages, discussions). We created explicit decision tables and exception forms so the automation could be deterministic.

Change Management

Staff were understandably protective of a process that protects federal compliance. Continuous demos and co-design sessions built trust and ownership.

Personal, Managerial & Technical Perspectives

Personal

Seeing the real impact of automation on people's daily work was deeply motivating. The project reinforced that technology only succeeds when it respects the humans who use it.

Managerial

Using every meeting as a collaboration opportunity and treating feedback as a gift produced far better outcomes than defending early designs. Psychological safety accelerated iteration.

Technical

Low-code platforms still demand rigorous requirements, error handling, and testing discipline. The difference between a demo and a production system is in the edge cases and logging.

How This Project Helped Us Grow as IS Professionals

01

Requirements Discipline

We learned that incomplete or ambiguous requirements are the root of most downstream pain. Investing early pays exponential returns.

02

Stakeholder Empathy

Walking the Gemba and co-designing with end users transformed us from “solution deliverers” into trusted partners.

03

Low-Code Maturity

Mastering Power Platform at production quality—error handling, logging, approval patterns—raised our capability ceiling.

04

Communication Craft

Translating complex automation logic into clear visuals and training materials became a core professional skill.

Recommended Next Steps

1

Pilot & Measure

Run the automation in parallel with the manual process for 2–3 weeks and quantify time savings and accuracy.

2

Training Rollout

Deliver hands-on sessions for all Registrar staff who will interact with the SharePoint approval interface.

3

Exception Form Expansion

Grow the shared exception form to cover more edge cases discovered during pilot.

4

Dashboarding

Add a Power BI or SharePoint dashboard so leadership can see weekly compliance trends at a glance.

5

Handover Package

Finalize technical runbooks, flow diagrams, and support contacts so RMU IT can own the solution long-term.

Thank You

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Sooyoung Lee · Robert Bost · Paul Arnold

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